Aufg.: 1a

$$\vec{B} = (0,0,I) \times \frac{(x,y,0)}{(x^2+y^2)^{3/2}} = \frac{(\overset{1}{-Iy}, \overset{2}{Ix}, 0)}{(x^2+y^2)^{3/2}} = I\frac{(-y,x,}{(x^2+y^2)}$$

$$\mathrm{rot}\,\vec{B} = \vec{\nabla} \times \vec{B}$$

$$= (\overset{1}{\partial_y B_3 - \partial_z B_2}, \overset{2}{\partial_z B_1 - \partial_x B_3}, \overset{3}{\partial_x B_2 - \partial_y B_1})$$

$$\cancel{= (-\partial_z B_2, \partial_z B_1, \partial_x B_2 \; \partial_y B_1)}$$

$$= \left(0, 0, \frac{2(x^2+y^2)^{3/2} - 3x^2(\;)^{1/2} - 3y^2(\;)^{1/2}}{(x^2+y^2)^3}\right)$$

$$= \left(0, 0, \frac{2x^2+2y^2-3x^2-3y^2}{(x^2+y^2)^{5/2}}\right) = -\left(0, 0, \frac{1}{(x^2+y^2)^{3/2}}\right)$$

1b) $\mathrm{div}\,\vec{B} = \vec{\nabla} \cdot \vec{B}$

$$= \partial_x B_1 + \partial_y B_2 + \partial_z B_3$$

$$= \frac{3}{4} I \frac{yx}{(x^2+y^2)^{5/2}} - \frac{3}{4} I \frac{xy}{(x^2+y^2)^{5/2}} = 0$$

1c) Nein da $\mathrm{rot}\,\vec{B} \neq 0$

Aufgabe 2

a) $\operatorname{rot}(x\sin z,\ z\ln y,\ y)$

$$= \nabla \times (\qquad\qquad) = \Big(\underbrace{\partial_y y}_{2\ 3} - \overset{1}{\partial_z}(z\ln y),\ \underset{1\ 3}{\partial_z}(x\sin z) - \overset{2}{\partial_x y},\ \underset{1\ 2}{\partial_x}(z\ln y) - \overset{1}{\partial_y}(x\sin z)\Big)$$

$$= (1-\ln y,\ x\cos z,\ 0)$$

b) $\operatorname{grad}(\vec{A}\cdot\vec{r}) = \operatorname{grad}(ax+by+cz)$

$$= (a,\ b,\ c)$$

c) $\operatorname{div}\operatorname{grad} U = \vec{\nabla}\cdot(U_{,x},\ U_{,y},\ U_{,z})$

$$= U_{,xx} + U_{,yy} + U_{,zz}$$

$$= \left(-\frac{1}{2}Q\frac{2x}{r^3}\right)_{,x} + \cdots$$

$$= -Q\left[\left(\frac{x}{r^3}\right)_{,x} + \left(\frac{y}{r^3}\right)_{,y} + \left(\frac{z}{r^3}\right)_{,z}\right]$$

$$= -Q\frac{1}{r^6}\Big[\underbrace{3r^3 - 3x^2r - 3y^2r - 3z^2r}_{3r^3-3r^3}\Big] = 0$$

Aufgabe 3

a)
$$y' = \frac{1 + c\frac{y}{x}}{c - \frac{y}{x}}$$

Nenn weg Bruch $\frac{y}{x} \neq c$, $x \neq 0$

$$z = \frac{y}{x} \qquad y' = (xz)' = z + xz'$$

$$z + xz' = \frac{1 + cz}{c - z}$$

$$xz' = \frac{1 + \cancel{cz} - \cancel{cz} + z^2}{c - z}$$

$$\frac{c - z}{1 + z^2}\, dz = \frac{1}{x}\, dx$$

$$\int \frac{c - z}{1 + z^2}\, dz = \ln x + C$$

$$c \cdot \operatorname{arctg} z - \frac{1}{2}\ln(1 + z^2) = \ln x + C$$

$$2c \operatorname{arctg} z - 2\ln x + C^* = \ln(1 + z^2)$$

$$c \operatorname{arctg} z - \ln x + C = \ln(1 + z^2)^{\frac{1}{2}}$$

$$C^* e^{c \operatorname{arctg} z} \frac{1}{x} = \sqrt{1 + z^2} = \sqrt{1 + \frac{y^2}{x^2}}$$

$$\| \; C^* e^{c \operatorname{arctg} \frac{y}{x}} = \sqrt{x^2 + y^2} \; \|$$

b)

$C^* = 1$

$r = e^{c\varphi}$

$r = e^{c \operatorname{arctg} \frac{y}{x}}$ eingesetzt

Aufgabe 4

a) $y' + \frac{2x}{1+x^2} y = \frac{2x^2}{1+x^2}$

linear

kein $y = 1c$.

$$\frac{dy}{dx} = -\frac{2x}{1+x^2} y$$

$$\frac{dy}{y} = -\frac{2x}{1+x^2} dx$$

$$\ln y = -\int \frac{2x}{1+x^2} dx = -\ln(1+x^2) + C.$$

$$y = \frac{1}{1+x^2} C^*$$

$$y' = -\frac{2x}{(1+x^2)^2} C^* = -\frac{2x}{1+x^2} y$$

allg. Lsg.:

$$y = C \frac{1}{1+x^2} + \frac{2}{3} x^3 \frac{1}{1+x^2}$$

b) Inhom. durch Variation der Konstant.

$$y^{(p)} = C(x) \frac{1}{1+x^2}$$

$$\Rightarrow C' \frac{1}{1+x^2} = \frac{2x^2}{1+x^2} \qquad + y_H' + \frac{2x}{1+x^2} y_H$$

$$\Rightarrow C' = 2x^2 \Rightarrow C = \frac{2}{3}x^3$$

Probe:

$$y' = \cancel{\frac{-2x}{(1+x^2)^2}C} + 2\frac{x^2}{1+x^2} \cancel{- \frac{2}{3}x^3 \frac{2x}{(1+x^2)^2}}$$

$$\frac{2x}{1+x^2}y = \cancel{\frac{2x}{(1+x^2)^2}C} \cancel{+ \frac{4}{3}x^4 \frac{1}{(1+x^2)^2}}$$

$$= 2\frac{x^2}{1+x^2}$$

Aufgabe 5

von $\sin\phi$ (Polynome in ϕ

$$\sin\phi \approx 0 + \cos\phi\Big|_{\phi=0}\phi - \frac{1}{2}\sin\phi\Big|_{\phi=0}\phi^2 - \frac{1}{6}\cos\phi\Big|_{\phi=0}\phi^3$$

$$\approx \phi - \frac{1}{6}\phi^3$$

bei der ersten

$$\ddot{\phi} + \frac{g}{l}\phi = 0$$

$$\phi = C e^{\lambda t}$$

$$\lambda^2 + \frac{g}{l} = 0 \qquad \frac{g}{l} = \omega^2$$

$$\lambda_{1/2} = \pm i\omega$$

$$\phi = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$$

$$\phi(t=0) \overset{!}{=} \phi_0 \quad ; \quad C_1 + C_2 = \phi_0$$

$$\dot{\phi}(t=0) = i\omega(C_1 - C_2) = 0$$

$$C_1 = C_2$$

$$\phi = \frac{1}{2}\phi_0\left(e^{i\omega t} + e^{-i\omega t}\right)$$

$$= \phi_0 \cos\omega t$$

angezeigt!

Aufgabe 6

$$y''' - 5y'' + 8y' - 4y = x$$

$$y = c\,e^{\lambda x}$$

$$\lambda^3 - 5\lambda^2 + 8\lambda - 4 = 0$$

$$\lambda_1 = 1$$

$$(\lambda^3 - 5\lambda^2 + 8\lambda - 4) : (\lambda - 1 = \lambda^2 - 4\lambda + 4$$

$$1 - 5 + 8 - 4 = 0$$

$$-(\lambda^3 - \lambda^2)$$

$$(-4\lambda^2 + 8\lambda - 4)$$

$$-(-4\lambda^2 + 4\lambda)$$

$$(4\lambda - 4)$$

$$-(4\lambda - 4)$$

$$\emptyset$$

$$\lambda^2 - 4\lambda + 4 = 0$$

$$\lambda_{2/3} = 2 \pm \sqrt{4-4}$$

$$\lambda_{2/3} = 2$$

$$y = C_1 e^x + (C_2 + C_3 x)\, e^{2x}$$

$$y^{(p)} = A + Bx \qquad y^{(p)'} = B \qquad y^{(p)''} = 0$$

eingesetzt:

$$8B - 4A - 4Bx = \cancel{0}$$

$$8B - 4A = 0$$

$$-4B = 1 \quad \Rightarrow \quad B = -\frac{1}{4}$$

$$-2 = 4A \qquad A = -\frac{1}{2}$$

$$\boxed{y^{(p)} = -\frac{1}{2} - \frac{1}{4}x}$$

$$y = C_1 e^x + (C_2 + C_3 x)\, e^{2x} - \frac{1}{2} - \frac{1}{4}x$$

Aufgabe 7

a) $y_1' = y_1 - y_2 \longrightarrow y_2 = y_1 - y_1'$

$\cancel{y_2 = y_1' - y}$

$y_2' = 4y_1 - 3y_2 \qquad y_2' = y_1' - y_1''$

$\cancel{y_1' - y_1''} = 4y_1 - 3y_1 + \cancel{3y_1'}$

$\Rightarrow -y_1'' = 2y_1' + y_1$

$y_1'' + 2y_1' + y_1 = 0$

$y_1 = C e^{\lambda x}$

$\lambda^2 + 2\lambda + 1 = 0$

$\lambda_{1/2} = -1 \pm 0$

$$\boxed{y_1 = (c_{11} + c_{12} x) e^{-x}}$$

$y_2 = (c_{11} + c_{12} x) e^{-x} - c_{12} e^{-x} + (c_{11} + c_{12} x) e^{-x}$

$$\boxed{y_2 = (2c_{11} - c_{12} + 2c_{12} x) e^{-x}}$$

$-y_1'' = 4y_1' + 3y_1$

$y_1'' = 2y_1' + y_1$

$\boxed{c_{11} = 1}$

$y_1(0) \overset{!}{=} 1 \Rightarrow$ ~~$c_{11} = 1$~~ ~~$2c_{11} - c_{12} = 1 \Rightarrow c_{12} = 1$~~

$y_2(0) = 1 \Rightarrow 2 - c_{12} = 1 \Rightarrow \boxed{c_{12} = 1}$

$\Rightarrow \boxed{y_1 = (1+x) e^{-x} \quad ; \quad y_2 = (1+2x) e^{-x}}$

Aufgabe 7b

$$y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \qquad y' = \overbrace{\begin{pmatrix} 1 & -1 \\ 4 & -3 \end{pmatrix}}^{A} y$$

$$y = \vec{c}\, e^{\lambda x}$$

$$\lambda y = A y$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & -1 \\ 4 & -3-\lambda \end{vmatrix} = 0$$

$$-(1-\lambda)(3+\lambda) + 4 = 0$$

$$\Rightarrow -3 - \lambda + 3\lambda + \lambda^2 + 4 = 0$$

$$\Rightarrow \lambda^2 + 2\lambda + 1 = 0$$

$$\lambda_{1/2} = -1 \pm 0$$

$$y = (\vec{C}_1 + \vec{C}_2 x)\, e^{-x}$$

$$y(0,0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = (\vec{C}_1) = \begin{matrix} 1 \\ 1 \end{matrix}$$

$$y_1 = (1 + c_{21} x)\, e^{-x}$$

$$y_2 = (1 + c_{22} x)\, e^{-x}$$

noch unbekannt

[illegible]